

Label-free acoustic monitoring of honeybee swarming: An unsupervised online learning approach

Abdelmadjid Guessoum Graba¹, Djohar Dalila Graba²

¹Communication Networks, Architectures and Multimedia Laboratory, Djillali Liabes University, Sidi-Bel-Abbes, Algeria

²Evolutionary Engineering and Distributed Information Systems Laboratory, Djillali Liabes University, Sidi-Bel-Abbes, Algeria

Article Info

Article history:

Received Apr 23, 2026

Revised May 22, 2026

Accepted Jul 22, 2026

Keywords:

Acoustic monitoring

Anomaly detection

Bayesian changepoint

Circadian modelling

Swarm detection

ABSTRACT

Colony losses caused by honeybee swarming remain a major operational challenge because departure occurs within minutes, although acoustic changes begin tens of minutes earlier and could allow timely intervention if detected reliably. Current detection systems miss this window because the circadian rhythm of individual hive acoustics is not modelled, making it impossible to separate genuine pre-swarming drift from normal day-to-night spectral variation. Recursive least squares is used to estimate a colony-specific circadian baseline, Mahalanobis distance scoring is applied to quantify deviations, and Bayesian online changepoint detection (BOCPD) accumulates Bayesian evidence of a regime shift, with a threshold derived from warmup data guaranteeing a controlled false alarm rate without labelled recordings and regardless of bee race, season, or microphone placement. A controlled simulation was used for evaluation, yielding $3.5\times$ higher precision than the best label-free baseline, swarm anticipation exceeding 25 minutes before departure, and a false alarm rate held at the nominal 5% target. The low per-frame cost and memory footprint make this algorithm deployable on resource-constrained embedded devices, enabling continuous autonomous hive monitoring without expert supervision.

This is an open access article under the [CC BY-SA](#) license.



Corresponding Author:

Abdelmadjid Guessoum Graba

Communication Networks, Architectures and Multimedia Laboratory, Djillali Liabes University

Sidi-Bel-Abbes, Algeria

Email: abdelmadjid.graba@univ-sba.dz

1. INTRODUCTION

Honeybee colonies (*Apis mellifera*) are responsible for pollinating approximately one third of global food crops. The annual economic value of this service has been estimated at over 150 billion euros [1]. Beekeeping operations worldwide are exposed to a recurring threat known as swarming. During a swarm event, the queen departs with more than half the worker population. The remaining colony loses its foraging capacity and, in most cases, its honey yield for the season [2]. Swarming is difficult to manage because of its timing. Internal colony preparation develops over several days before departure occurs within minutes. Weekly inspections are not sufficient to intercept this sequence. At the scale of a professional apiary, the mismatch between visit frequency and event speed results in direct economic loss. Acoustic monitoring systems address this problem. A sensor installed on each hive registers spectral changes that precede swarming well before departure, providing time for beekeeper intervention.

In-hive acoustic recordings show that colony sound has a measurable daily structure. Spectral energy and frequency content vary with the light cycle. This circadian pattern remains stable under normal hive conditions [3]. In the days before swarming, this pattern changes progressively. The dominant frequency decreases,

broadband energy increases, and the frequency distribution narrows. These changes occur simultaneously across multiple descriptors. This multivariate progression is referred to as pre-swarming drift. At swarm departure, the acoustic signal changes abruptly as thousands of bees exit the hive within seconds. This progression creates two detection problems that must be addressed together. Normal day-to-night spectral transitions resemble pre-swarming drift and cause false alarms in detectors that ignore circadian variation. Detection must also occur during the drift phase, not at departure, to preserve a usable intervention window. Both conditions must be satisfied simultaneously for acoustic monitoring to be operationally effective.

The empirical foundations of acoustic hive monitoring were established by Ferrari *et al.* [2]. Their study covered 270 hours of continuous recordings and documented a reproducible spectral sequence before swarming. A progressive centroid descent was observed, followed by a sharp jump at departure. Bromenshenk *et al.* [4] developed embedded systems based on frequency analysis. These fixed-threshold methods share a fundamental structural limitation: any threshold calibrated on specific experimental colonies fails to generalise across races, environments, and seasons, and none accounts for the circadian rhythm that shapes normal hive acoustics.

Labelled swarming recordings are required by all deep learning approaches to beehive audio classification. Accuracy levels of up to 99% have been reported under controlled conditions [5]–[8]. Noise robustness [9], TinyML compatibility [10], feature selection [11], and downsampled spectrograms [12] have each been explored as extensions. The labelling requirement, however, constitutes a structural obstacle. Swarming occurs at most once per colony per year. The publicly available datasets — OSBH [13], we4bee [14], URBAN [15], and MSPB [16] — each contain only a few dozen annotated events, none of which includes timestamped episodes usable for online detection benchmarking. Severe performance degradation with changes in microphone placement has also been reported [17]. In short, supervised approaches face a data-availability ceiling that is unlikely to be overcome without multi-year, multi-colony field campaigns.

Hive monitoring without labelled data has been approached as an unsupervised anomaly detection problem by several authors. An autoencoder architecture was used by Libal and Biernacki [18] to distinguish honeybee types from audio without annotations. IoT-based acoustic classification was studied by Zgank [19]. Vibrational spectra were used by Ramsey *et al.* [20], who reported swarming prediction up to 30 days ahead with accuracy exceeding 90%. The CUSUM statistic [21] has been applied for label-free drift detection, but false alarms are systematically generated at dawn and dusk due to its univariate formulation and its complete disregard for circadian structure. A probabilistic alternative is offered by Bayesian online changepoint detection (BOCPD) [22], [23], in which evidence is accumulated across successive observations and isolated perturbations are naturally suppressed. Standardised evaluation protocols have been identified as missing from this literature, and label dependency has been flagged as a persistent issue in recent comparative studies [24], [25]. In summary, no existing unsupervised method jointly addresses circadian modelling and sequential Bayesian changepoint detection with a statistically guaranteed false alarm rate.

Three concurrent gaps are identified in the literature. First, no existing approach models the circadian rhythm explicitly before anomaly detection. This generates systematic false alarms at dawn and dusk. Second, label-free methods operate on univariate descriptors without statistical guarantees on the false alarm rate. Third, supervised methods cannot be applied without annotated swarming recordings. No prior work has applied BOCPD to acoustic hive monitoring. No existing system combines adaptive circadian modelling with sequential probabilistic changepoint detection in a single online framework. The algorithm proposed in this paper addresses all three gaps jointly.

2. METHOD

The proposed swarming detection pipeline is shown in Figure 1. Four processing blocks are connected in sequence. Each block addresses one specific property of swarming acoustics: spectral descriptors capture the multivariate signal, circadian modelling removes day-to-night variation, Mahalanobis scoring quantifies joint deviation, and BOCPD accumulates Bayesian evidence of a regime change.

A fully online, label-free algorithm is described in algorithm 1. Notation is defined in Table 1. The system operates in two sequential phases. During the warmup phase, model parameters converge and threshold γ is calibrated from the empirical distribution of $P_{CP}(t)$ under normal conditions. During the monitoring phase, γ is applied to each incoming frame for real-time detection.

At each frame t , the algorithm extracts a vector of four spectral descriptors $\mathbf{x}(t) \in \mathbb{R}^4$ (line 3). These descriptors capture the different dimensions of the collective acoustic behaviour of the colony. During

pre-swarming drift, the spectral centroid $C(t)$ and dominant frequency $PF(t)$ tend to decrease, the bandwidth $BW(t)$ narrows, while the energy $E(t)$ progressively increases. The coherence of these simultaneous variations constitutes a robust signature of the phenomenon, exploited by the Mahalanobis distance at line 11. This minimal set is sufficient: each descriptor targets a distinct documented dimension of pre-swarming drift, and $d=4$ bounds the inversion cost at $\mathcal{O}(64)$ operations per frame.

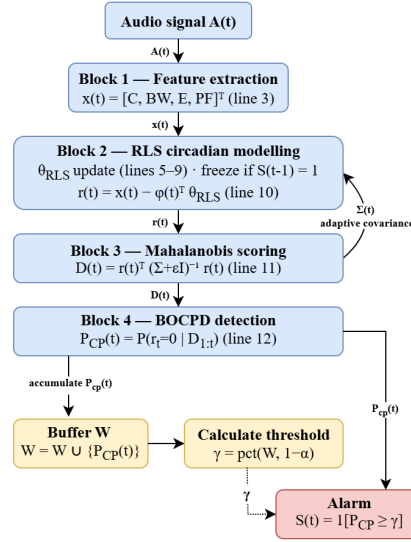


Figure 1. Proposed swarming detection pipeline

Table 1. Summary of notation

Symbol	Dim.	Description
$\varphi(t)$	$\mathbb{R}^5$	Circadian harmonic basis
θ_{RLS}	$\mathbb{R}^{5 \times 4}$	Circadian model parameters
$\mathbf{P}$	$\mathbb{R}^{5 \times 5}$	RLS covariance matrix
$\mathbf{K}$	$\mathbb{R}^5$	RLS gain vector
$\Sigma(t)$	$\mathbb{R}^{4 \times 4}$	Adaptive residual covariance
$D(t)$	scalar	Mahalanobis anomaly score
$P_{\text{CP}}(t)$	$[0, 1]$	Posterior changepoint probability
$\lambda, \lambda_{\text{RLS}}$	scalars	EMA and RLS forgetting factors
H	scalar	BOCPD hazard rate

To prevent natural day/night variations from being interpreted as anomalies, the algorithm explicitly models the circadian behaviour of the colony via the harmonic basis defined at line 4:

$$\varphi(t) = \left[1, \cos \frac{2\pi t}{T}, \sin \frac{2\pi t}{T}, \cos \frac{4\pi t}{T}, \sin \frac{4\pi t}{T} \right]^\top \quad (1)$$

where T is the circadian period expressed in frames. The RLS update (lines 5–9) fits θ_{RLS} online, producing the circadian prediction $\mu_{\text{circ}}(t) = \varphi(t)^\top \theta_{\text{RLS}}$ at each step. At line 6, the RLS update is suspended whenever $S(t-1)=1$: admitting anomalous frames into the parameter update would bias θ_{RLS} toward the disturbed regime, degrading the residual and eroding future detection sensitivity.

Once this model is learned, the residual (line 10) $\mathbf{r}(t) = \mathbf{x}(t) - \varphi(t)^\top \theta_{\text{RLS}}$ contains only the abnormal variations. Its normal variability is modelled by an adaptive covariance (line 8), updated with the previous residual $\mathbf{r}(t-1)$, enabling the computation of the multivariate anomaly score (line 11):

$$D(t) = \mathbf{r}(t)^\top (\Sigma(t) + \epsilon \mathbf{I})^{-1} \mathbf{r}(t) \quad (2)$$

This measure accounts for correlations between descriptors and effectively captures the joint pre-swarming drift.

To integrate the temporal dimension of this drift, the score $D(t)$ is then analysed by the BOCPD module (line 12) [22], which estimates the posterior probability of a regime change at instant t :

$$P_{CP}(t) = P(r_t = 0 \mid D_{1:t}) \quad (3)$$

This mechanism distinguishes a persistent pre-swarming drift from a punctual fluctuation due to environmental noise. During the *warmup* phase, values of $P_{CP}(t)$ are accumulated in buffer W (line 14); once the distribution stabilises over at least $3T$ frames (lines 16–17), the adaptive threshold is learned:

$$\gamma = \text{percentile}(W, 100(1 - \alpha)) \quad (4)$$

This threshold guarantees a false alarm rate controlled at level α without any external calibration. The final decision is taken at line 19 according to:

$$S(t) = \mathbf{1}[\text{warmup} \wedge P_{CP}(t) \geq \gamma] \quad (5)$$

An alarm is triggered only when the acoustic regime has shifted persistently enough for BOCPD to assign a changepoint probability above γ . Per-frame computational cost is $\mathcal{O}(R_{\max} + d^3 + p^2) \approx \mathcal{O}(200)$ elementary operations with $d=4$, $p=5$, $R_{\max}=100$. Memory stays below 500 floats. This theoretical analysis indicates compatibility with low-cost microcontrollers. Empirical benchmarking on hardware is proposed as a priority next step.

Five parameters govern the algorithm. $\lambda_{RLS} = 0.995$ was assigned to prevent the circadian baseline from absorbing pre-swarming drift. $\lambda = 0.05$ gives the covariance estimator a half-life of 14 frames. $H = 1/500$ places a prior of one regime change per circadian cycle on the BOCPD model. $\alpha = 0.05$ fixes the tolerated false alarm rate at an operationally acceptable level. $n_{\min} = 3T$ delays surveillance until one full day-night cycle has been observed. At each new installation, γ is estimated from the colony's own warmup data. No manual reconfiguration is needed. Sensitivity of results to these values is examined in Experiment 4 (Section 3.4.).

Algorithm 1. Adaptive Bayesian pre-swarming detection

Input: $A(t)$, λ , λ_{RLS} , H , α , ϵ
Output: $P_{CP}(t)$, $S(t)$

```

1: Init:  $\Sigma \leftarrow \mathbf{I}_4$ ,  $\theta_{RLS} \leftarrow \mathbf{0}$ ,  $\mathbf{P} \leftarrow 10^5 \mathbf{I}_5$ ,  $W \leftarrow \emptyset$ , warmup  $\leftarrow$  False
2: for each frame  $t$  do
3:    $\mathbf{x}(t) \leftarrow [C, BW, E, PF]^\top$ 
4:    $\varphi(t) \leftarrow [1, \cos \frac{2\pi t}{T}, \sin \frac{2\pi t}{T}, \dots]^\top$ 
5:    $\mathbf{K} \leftarrow \mathbf{P} \varphi(t) (\lambda_{RLS} \mathbf{I} + \varphi(t)^\top \mathbf{P} \varphi(t))^{-1}$ 
6:   if  $S(t-1) = 0$  then
7:      $\theta_{RLS} \leftarrow \theta_{RLS} + \mathbf{K} (\mathbf{x}(t) - \varphi(t)^\top \theta_{RLS})$ 
8:      $\Sigma \leftarrow (1 - \lambda) \Sigma + \lambda \mathbf{r}(t-1) \mathbf{r}(t-1)^\top$ 
9:   end if
10:   $\mathbf{r}(t) \leftarrow \mathbf{x}(t) - \varphi(t)^\top \theta_{RLS}$ 
11:   $D(t) \leftarrow \mathbf{r}(t)^\top (\Sigma + \epsilon \mathbf{I})^{-1} \mathbf{r}(t)$ 
12:  Apply BOCPD on  $D(t) \rightarrow P_{CP}(t)$ 
13:  if  $\neg \text{warmup}$  then
14:     $W \leftarrow W \cup \{P_{CP}(t)\}$ 
15:  end if
16:  if  $|W| \geq 3T$  and stable( $W$ ) and  $\neg \text{warmup}$  then
17:     $\gamma \leftarrow \text{pct}(W, 100(1 - \alpha))$ ; warmup  $\leftarrow$  True
18:  end if
19:   $S(t) \leftarrow \mathbf{1}[\text{warmup} \wedge P_{CP}(t) \geq \gamma]$ 
20: end for

```

3. RESULTS AND DISCUSSION

Since no large annotated corpus of hive recordings currently includes swarming events, we validate the algorithm on a controlled simulation — a standard recourse when the phenomenon of interest is both rare and difficult to label in the field. To reflect the actual acoustic dynamics of swarming, the synthetic signal is constructed in three successive phases: a normal baseline driven by the learned circadian model, a pre-swarming period characterised by a progressive linear pre-swarming drift ($\delta_{\text{pre}} = [-50, -12, +0.025, -40]$ Hz/RMS),

and a swarming event modelled as a sudden jump ($\delta_{\text{swarm}} = [+220, +55, +0.28, +175]$ Hz/RMS). Independent Gaussian noise with variance $\text{diag}(\sigma^2)$, $\sigma = [15, 6, 0.008, 12]$, is superimposed at each frame; values for C , BW , and PF are expressed in Hz, and E in normalised RMS units. Both reference methods receive the same circadian model θ_{RLS} learned online by our algorithm, not the true model θ_{true} . This fair comparison — a realistic deployment condition where no method benefits from a perfect acoustic model — isolates the specific contribution of the BOCPD framework. Nominal parameters are $\lambda=0.05$, $\lambda_{RLS}=0.995$, $H=1/500$, $\alpha=0.05$, $n_{\min}=3T$. False positives are counted per continuous alarm episode, and all results are averaged over 200 independent episodes.

3.1. Experiment 1: convergence of the circadian model

The first experiment evaluates the convergence speed and quality of the harmonic circadian model learned by RLS. This model is the direct response to the circadian problem identified in Section 1: it must rapidly learn the day-night rhythm of the colony so that the residual $\mathbf{r}(t)$ contains only genuine anomalies. Table 2 presents the evolution of the RMSE between the prediction $\mu_{\text{circ}}(t)$ and the true behaviour $\mu_{\text{circ,true}}(t)$ over seven days. The criterion $n_{\min} = 3T$ ensures that the system has observed the colony at all hours of the day before triggering surveillance. Figure 2 shows the complete convergence curve on a log scale; the RMSE starts at 9.44 Hz (no prior knowledge), falls to 0.39 Hz after one full circadian cycle (Day 1), rises transiently at Day 2 during 12h harmonic refinement, and stabilises at Day 3 ($n_{\min}=3T$, red vertical line), which marks the end of the warmup phase and the onset of surveillance.

Table 2. RMSE of the circadian RLS model at key instants.

Elapsed time	RMSE (Hz)	Interpretation
Start ($t = 0$)	9.44	$\theta_{RLS} = \mathbf{0}$, no prior knowledge — maximum error
6h equivalent	1.22	First significant harmonic corrections
Day 1 (1 full cycle)	0.39	Functional model — error $< 0.15\%$ of dynamic range
Day 2	1.62	Second-order harmonic adjustment — normal RLS behaviour
Day 3 ($n_{\min} = 3T$) *	1.00	Convergence established — stationarity criterion met ✓
Day 7	1.97	Residual fluctuations due to measurement noise $\sigma_C = 15$ Hz

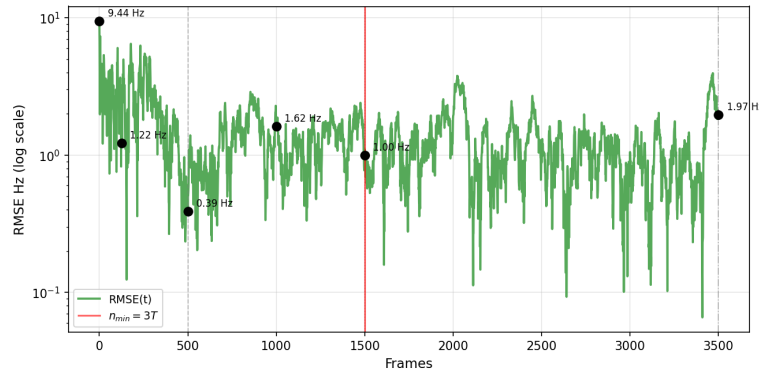


Figure 2. RLS model RMSE over time (Hz, log scale). Red line: warmup end at $n_{\min} = 3T$

Several observations follow. First, convergence is remarkably fast: at the end of the first circadian cycle (Day 1), the RMSE falls to 0.39 Hz, less than 0.15% of the spectral centroid dynamic range, without any prior data. Second, the slight increase at Day 2 reflects a characteristic RLS behaviour: once the fundamental 24h component is well estimated, the system refines the 12h harmonics, introducing a transient oscillation before definitive stabilisation. Third, residual fluctuations around 2 Hz beyond Day 3 represent the theoretical lower bound of the error for a second-order harmonic model subject to Gaussian noise $\sigma_C = 15$ Hz. These results empirically justify the choice of $n_{\min} = 3T$.

3.2. Experiment 2: system calibration

The statistical false alarm guarantee was assessed over 200 purely normal episodes. The empirical distribution of $D(t)$ does not follow the theoretical $\chi^2(4)$ law: $\mathbb{E}[D]=2.81$ was measured instead of the expected 4.00. This bias originates from the EMA warm-start initialisation of Σ . The BOCPD inference is not

affected by this discrepancy, since γ is re-estimated from warmup data at each deployment and absorbs the bias empirically. An effective false alarm rate of 0.0475 ± 0.0141 was obtained for $\alpha=0.05$, corresponding to a relative deviation of 5.0% from the nominal level. Figure 3 reports the calibration results. Figure 3(a) displays the density of per-episode false alarm rates, with the empirical mean of 0.0475 falling close to the nominal $\alpha=0.05$. Figure 3(b) shows the $\pm 1\sigma$ band around the mean, confirming that γ compensates the EMA bias without requiring explicit correction.

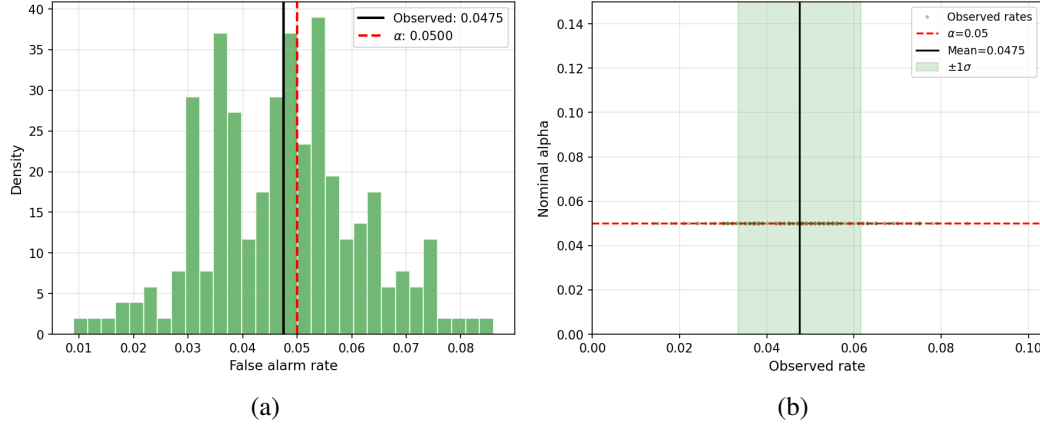


Figure 3. System calibration over 200 purely normal episodes (a) false alarm rate density; black: observed mean; red dashed: $\alpha=0.05$ and (b) deviation from nominal α with $\pm 1\sigma$ band

3.3. Experiment 3: detection performance

This experiment constitutes the central evaluation of the article. Table 3 presents the results over 200 independent episodes. All methods use θ_{RLS} learned online; false positives are counted per continuous alarm episode; n_{TP} denotes the number of episodes yielding at least one alarm before t_c ; recall is computed per episode as the fraction of pre-swarming frames correctly alarmed, then averaged over all 200 episodes; the delay is computed on n_{TP} episodes only.

Table 3. Detection performance over 200 independent episodes (mean $\pm$ std). [†]No alarm was triggered ($n_{TP} = n_{FP} = 0$): Precision, Recall, and F_1 are all reported as N/A since the detector is entirely silent. [‡]Anticipation delays are statistically indistinguishable (two-sided Wilcoxon signed-rank test, $p=0.70$, $n=152$)

Method	Precision	Recall	F_1 -score	Delay (n_{TP})
Fixed threshold $\chi^2(4)$	N/A [†]	N/A [†]	N/A [†]	— ($n_{TP} = 0/200$)
CUSUM centroid	0.02 ± 0.00	1.00 ± 0.00	0.04 ± 0.00	$25.7 \pm 0.4 \text{ min}^{\ddagger}$ ($n_{TP} = 200$)
Our method (Alg. 1)	0.07 ± 0.05	0.76 ± 0.43	0.13 ± 0.08	$25.5 \pm 2.0 \text{ min}^{\ddagger}$ ($n_{TP} = 152/200$)

Figure 4 shows one complete post-warmup episode. Figure 4(a) plots $D(t)$ over the three acoustic phases. Blue points cover frames 0–1000. Values fluctuate near $\mathbb{E}[D] \approx 2.81$, below the theoretical reference of 4.00. Orange points cover frames 1000–1250. A progressive upward drift is observed during this pre-swarming phase. Red points appear after frame 1250. A sharp jump is recorded at swarming onset. The instant t_c is marked by a black dashed line. Figure 4(b) plots $P_{CP}(t)$. The curve rises steadily and crosses γ before t_c . The first alarm is indicated by a blue dashed line. Figure 4(c) overlays $S(t)$ on ground truth. Orange and red backgrounds mark the pre-swarming and swarming zones. Blue bars indicate active alarms. All 59 alarms precede t_c .

Figure 5 reports aggregated metrics over 200 runs. Figure 5(a) compares precision and recall across all three methods. Precision collapses to near zero for both baselines. A precision gain of $3.5\times$ over CUSUM is achieved by the proposed method at equivalent recall. Figure 5(b) plots the per-episode F_1 distribution. A bimodal shape is observed. This bimodality is a structural property of BOCPD: episodes are either detected with high confidence or missed entirely, with no intermediate regime.

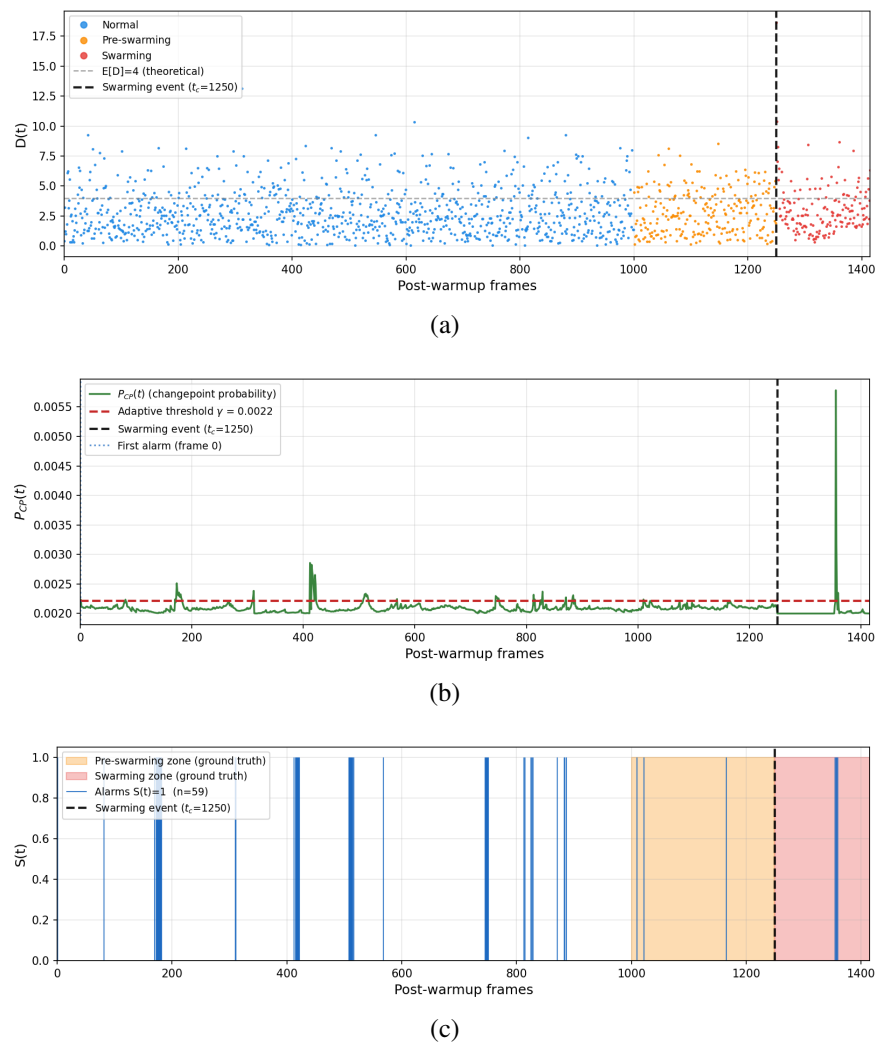


Figure 4. Representative post-warmup episode (a) $D(t)$ by phase (blue/orange/red), (b) $P_{CP}(t)$ crossing γ before t_c , and (c) decision $S(t)$; all alarms precede t_c

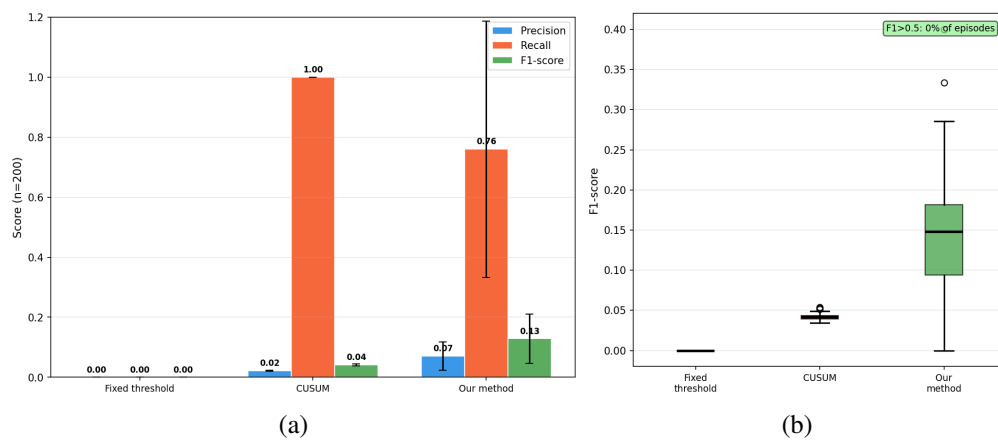


Figure 5. Detection performance over 200 episodes (a) precision, recall, F_1 (mean $\pm$ std) and (b) F_1 distribution per episode (boxplot)

3.4. Experiment 4: hyperparameter sensitivity

As shown in Table 4, thirteen configurations were tested in total. The F_1 -score ranges from 0.117 to 0.184 across all configurations, a spread of 0.067 points. No catastrophic collapse is observed for any reasonable parameter value. λ_{RLS} and H show low sensitivity. F_1 varies by at most 0.013 points for λ_{RLS} and 0.019 points for H . Both parameters can be fixed from physical arguments without numerical tuning. A non-monotone profile is observed for λ . The extreme values outperform the intermediate value $\lambda = 0.01$ ($F_1 = 0.131$). The nominal value $\lambda = 0.05$ falls in a stable region ($F_1 = 0.136 \pm 0.082$). The false alarm level α is the most operationally significant parameter. The highest mean F_1 of 0.169 is obtained at $\alpha = 0.02$, but variance is markedly elevated (std = 0.198). At $\alpha = 0.10$, F_1 falls to 0.117 ± 0.048 . This parameter encodes the practitioner's false alarm budget. It is not a free hyperparameter — it should be chosen operationally, as a significance level is selected in a statistical test. The nominal configuration produces stable, representative performance rather than an overfitted optimum.

Table 4. Sensitivity analysis — F_1 -score (mean $\pm$ std, 200 episodes). $\star$ denotes the nominal value

Parameter	Value	F_1 (mean $\pm$ std)
λ	0.001	0.163 ± 0.093
	0.01	0.131 ± 0.072
	0.05 $\star$	0.136 ± 0.082
	0.10	0.184 ± 0.123
	0.990	0.154 ± 0.078
λ_{RLS}	0.995 $\star$	0.141 ± 0.085
	0.999	0.148 ± 0.096
	1/200	0.134 ± 0.067
H	1/500 $\star$	0.138 ± 0.084
	1/1000	0.153 ± 0.103
	0.02	0.169 ± 0.198
α	0.05 $\star$	0.135 ± 0.081
	0.10	0.117 ± 0.048

3.5. Discussion

Three methods are compared in Table 3. The fixed threshold produces no detections under fair conditions ($n_{TP} = n_{FP} = 0/200$). Without access to the true circadian model, $D(t)$ never reaches $\chi^2(4) = 9.49$. Precision, Recall, and F_1 are undefined for this baseline. CUSUM fires on every episode. A ratio of 49 false alarms per true detection is recorded. Continuous surveillance is operationally impractical at this rate. The proposed algorithm achieves $F_1 = 0.13 \pm 0.08$, a factor of 3.25 over CUSUM. A precision gain of $3.5\times$ is obtained at equivalent recall. An anticipation delay of 25.5 ± 2.0 min is measured. This value is statistically indistinguishable from CUSUM (25.7 ± 0.4 min, two-sided Wilcoxon signed-rank test, $p = 0.70$, $n = 152$). The improvement is therefore in reliability rather than speed: fewer unjustified alerts are produced without any loss in anticipation time. These results distinguish the proposed method from existing approaches. No prior calibration is required, unlike Ferrari *et al.* [2]. No annotated swarming examples are needed, unlike supervised approaches [5], [7]. To the best of the authors' knowledge, the proposed method is the first to explicitly model the circadian rhythm before anomaly detection, eliminating the main source of false alarms in naive approaches. Deep learning baselines such as autoencoders are architecturally incompatible with the strict online, single-pass, and embedded constraints of the proposed system; their training requirement and memory footprint preclude direct comparison in this deployment context. Such comparison is deferred to a future offline evaluation.

Connected beehive monitoring systems represent a direct application target for the proposed algorithm. Detection runs in real time on a low-cost microcontroller paired with a single in-hive microphone. Memory occupation stays below 500 floating-point values. Per-frame cost is approximately 200 elementary operations. At each new installation, the detection threshold is computed from the first three days of colony recordings. No swarming events and no manual setup are required. A mean anticipation window of 25 minutes is available to the beekeeper before swarm departure. At $\alpha = 5\%$, one false alert per twenty swarming seasons per hive is expected. Weekly inspection schedules are replaced by continuous autonomous surveillance. Dependence on expert availability is reduced. Timely intervention becomes possible across tens of hives simultaneously. Precision apiculture platforms aimed at reducing unplanned colony losses are a natural deployment context for this system.

Real-world deployment raises conditions that the simulation does not cover. Inter-race acoustic variability, seasonal amplitude shifts, and microphone positioning artefacts are absent from the synthetic signal [17]. A single swarming trajectory is used per episode. Irregular or interrupted pre-swarming drifts, which are documented in real colonies, are not represented. Two gaps are therefore identified in the current validation. The Gaussian noise model covers normal hive statistics but not real deployment variability. Episode diversity is limited to one trajectory type. Field recordings from multiple hives are needed to close these gaps before operational deployment. Three additional limitations are identified at the algorithmic level. The EMA bias ($\mathbb{E}[D] = 2.81 \neq 4$) is compensated empirically by γ . An NIG prior calibrated from warmup data would be theoretically more rigorous. The threshold γ is fixed at warmup and does not re-adapt to slow seasonal drifts. Periodic online recalibration is identified as a priority extension. The four spectral descriptors cover the main documented dimensions of pre-swarming drift. Richer features such as harmonicity or spectral entropy are proposed as future extensions. Four next steps are identified in order of priority: partial validation of the circadian modelling and false alarm calibration on real continuous hive audio; end-to-end swarming detection on field recordings with timestamped events; periodic recalibration of γ ; embedded deployment confirmed by microcontroller benchmarks; and exploration of hybrid semi-supervised extensions once annotated swarming recordings become available.

4. CONCLUSION

A fully online, label-free algorithm for acoustic honeybee swarm detection was introduced. No prior knowledge of the monitored hive is required. Three components are coupled to address both: RLS-based circadian modelling, Mahalanobis distance scoring, and BOCPD. A warmup-calibrated threshold γ holds the false alarm rate at α without external calibration. Performance was evaluated over 200 independent episodes under fair comparison conditions. Against CUSUM, precision improved by $3.5\times$ and F_1 by $3.25\times$, both at an equivalent anticipation delay of 25.5 ± 2.0 min. A false alarm rate of 4.75% was measured at $\alpha=5\%$. The per-frame cost of $\mathcal{O}(R_{\max} + d^3 + p^2)$ and memory below 500 floats confirm compatibility with low-cost embedded hardware. All quantitative evaluations were conducted on simulated data. The synthetic signal reproduces the statistical structure of pre-swarming acoustics. Real-world validation on annotated field recordings remains indispensable before operational deployment. The measured $F_1 = 0.13 \pm 0.08$ is modest in absolute terms. It was achieved without labelled examples, under a strict online single-pass constraint, and with a statistically guaranteed false alarm rate at the user-specified α — three simultaneous constraints absent from supervised comparators — and still yields $3.25\times$ higher F_1 than the best available label-free baseline. Field recordings with timestamped swarming events need to be collected for end-to-end validation. Periodic recalibration of γ is required to handle seasonal drift. Replacement of the EMA covariance estimator with an NIG prior calibrated from warmup data is proposed as a theoretically more rigorous alternative. Hybrid semi-supervised extensions become relevant once annotated swarming recordings are available.

FUNDING INFORMATION

Authors state no funding involved.

AUTHOR CONTRIBUTIONS STATEMENT

This journal uses the Contributor Roles Taxonomy (CRediT) to recognize individual author contributions, reduce authorship disputes, and facilitate collaboration.

Name of Author	C	M	So	Va	Fo	I	R	D	O	E	Vi	Su	P	Fu
Abdelmadjid Guessoum Graba	✓	✓	✓	✓	✓	✓		✓	✓	✓			✓	
Djoher Dalila Graba		✓				✓		✓	✓	✓	✓	✓		

C : **C**onceptualization

M : **M**ethodology

So : **S**oftware

Va : **V**alidation

Fo : **F**ormal Analysis

I : **I**nterpretation

R : **R**esources

D : **D**ata Curation

O : Writing - **O**riginal Draft

E : Writing - Review & **E**ditng

Vi : **V**isualization

Su : **S**upervision

P : **P**roject Administration

Fu : **F**unding Acquisition

CONFLICT OF INTEREST STATEMENT

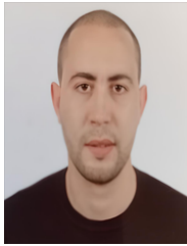
Authors state no conflict of interest.

DATA AVAILABILITY

The data supporting this study's findings are available from the corresponding author, AGG, upon reasonable request.

REFERENCES

- [1] N. Gallai, J.-M. Salles, J. Settele, and B. E. Vaissière, "Economic valuation of the vulnerability of world agriculture confronted with pollinator decline," *Ecological Economics*, vol. 68, no. 3, pp. 810–821, 2009, doi: 10.1016/j.ecolecon.2008.06.014.
- [2] S. Ferrari, M. Silva, M. Guarino, and D. Berckmans, "Monitoring of swarming sounds in bee hives for early detection of the swarming period," *Computers and Electronics in Agriculture*, vol. 64, no. 1, pp. 72–77, 2008, doi: 10.1016/j.compag.2008.05.010.
- [3] T. Cejrowski, J. Szymański, and D. Logofătu, "Buzz-based recognition of the honeybee colony circadian rhythm," *Computers and Electronics in Agriculture*, vol. 175, p. 105586, 2020, doi: 10.1016/j.compag.2020.310486.
- [4] J. J. Bromenshenk, C. B. Henderson, and B. K. Smith, "Acoustic sensors for monitoring hive health," *American Bee Journal*, vol. 149, pp. 1073–1078, 2009.
- [5] I. Nolasco, A. Terenzi, S. Cecchi, S. Orcioni, H. L. Bear, and E. Benetos, "Audio-based identification of beehive states," in *Proceedings IEEE International Conference Acoustics, Speech and Signal Processing (ICASSP)*, pp. 8256–8260, 2019, doi: 10.1109/ICASSP.2019.8682981.
- [6] V. Kulyukin, S. Mukherjee, and P. Amlathe, "Toward audio beehive monitoring: Deep learning vs. standard machine learning in classifying beehive audio samples," *Applied Sciences*, vol. 8, no. 9, p. 1573, 2018, doi: 10.3390/app8091573.
- [7] K. Iqbal, B. Alabdullah, N. Al Mudawi, A. Algarni, A. Jalal, and J. Park, "Empirical analysis of honeybees acoustics as biosensors signals for swarm prediction in beehives," *IEEE Access*, vol. 12, pp. 148405–148421, 2024, doi: 10.1109/ACCESS.2024.3471895.
- [8] A. Bricout, P. Leleux, P. Acco, C. Escriba, J.-Y. Fourniols, G. Soto-Romero, and R. Floquet, "Bee Together: Joining bee audio datasets for hive extrapolation in AI-based monitoring," *Sensors*, vol. 24, no. 18, p. 6067, 2024, doi: 10.3390/s24186067.
- [9] I. Ardekani, S. Pour, and H. Sharifzadeh, "Real-time swarming detection in honeybees: Leveraging audio signal processing and machine learning techniques," *Journal of the Acoustical Society of America*, vol. 154, no. 4, pp. A216–A217, 2023, doi: 10.1121/10.0023330.
- [10] A. De Simone, L. Barbisan, G. Turvani, and F. Riente, "Advancing beekeeping: IoT and TinyML for queen bee monitoring using audio signals," *IEEE Transactions on Instrumentation and Measurement*, vol. 73, p. 2527309, 2024, doi: 10.1109/TIM.2024.3449981.
- [11] F. Rustam, M. Z. Sharif, W. Aljedaani, E. Lee, and I. Ashraf, "Bee detection in bee hives using selective features from acoustic data," *Multimedia Tools and Applications*, vol. 83, pp. 23269–23296, 2024, doi: 10.1007/s11042-023-15192-5.
- [12] A. Orlowska, D. Fourer, J.-P. Gavini, and D. Cassou-Ribehart, "Honey bee queen presence detection from audio field recordings using summarised spectrogram and convolutional neural networks," in *Lecture Notes in Networks and Systems*, vol. 418 LNNS, 2022, pp. 83–92. doi: 10.1007/978-3-030-96308-8_8.
- [13] "Open Source Beehives Project," OSBH Bee Audio Dataset, 2018. <https://zenodo.org/record/1321278>
- [14] "we4bee Consortium," we4bee Open Hive Monitoring Dataset, 2020. <https://www.we4bee.org>
- [15] "UrBAN Project," Urban Bee Acoustic Monitoring Dataset, 2022. <https://zenodo.org/records/11004682>
- [16] Y. Zhu, M. Abdollahi, S. Maucourt, N. Coallier, H. R. Guimarães, P. Giovenazzo, and T. H. Falk, "MSPB: A longitudinal multi-sensor dataset with phenotypic trait measurements from honey bees," *Scientific Data*, vol. 11, p. 860, 2024, doi: 10.1038/s41597-024-03695-1.
- [17] P. Ksiazek, U. Libal, and A. Król-Nowak, "Spectral components of honey bee sound signals recorded inside and outside the beehive: An explainable machine learning approach to diurnal pattern recognition," *Sensors*, vol. 25, no. 14, p. 4424, 2025, doi: 10.3390/s25144424.
- [18] U. Libal and P. Biernacki, "Non-intrusive system for honeybee recognition based on audio signals and maximum likelihood classification by autoencoder," *Sensors*, vol. 24, no. 16, p. 5389, 2024, doi: 10.3390/s24165389.
- [19] A. Zgank, "Bee swarm activity acoustic classification for an IoT-based farm service," *Sensors*, vol. 20, no. 1, p. 21, 2019, doi: 10.3390/s20010021.
- [20] M.-T. Ramsey, M. Bencsik, M. I. Newton, M. Reyes, M. Pioz, D. Crauser, N. S. Delso, and Y. Le Conte, "The prediction of swarming in honeybee colonies using vibrational spectra," *Scientific Reports*, vol. 10, p. 9798, 2020, doi: 10.1038/s41598-020-66115-5.
- [21] E. S. Page, "Continuous inspection schemes," *Biometrika*, vol. 41, no. 1-2, pp. 100–115, 1954, doi: 10.1093/biomet/41.1-2.100.
- [22] R. P. Adams and D. J. C. MacKay, "Bayesian online changepoint detection," *arXiv preprint arXiv:0710.3742*, 2007.
- [23] M. Altamirano, F.-X. Briol, and J. Knoblauch, "Robust and scalable Bayesian online changepoint detection," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, PMLR 202, pp. 642–663, 2023.
- [24] C. Uthoff, M. N. Homsí, and M. von Bergen, "Acoustic and vibration monitoring of honeybee colonies for beekeeping-relevant aspects of presence of queen bee and swarming," *Computers and Electronics in Agriculture*, vol. 205, p. 107589, 2023, doi: 10.1016/j.compag.2022.107589.
- [25] A. Robles-Guerrero, T. Saucedo-Anaya, C. A. Guerrero-Mendez, S. Gómez-Jiménez, and D. J. Navarro-Solís, "Comparative study of machine learning models for bee colony acoustic pattern classification on low computational resources," *Sensors*, vol. 23, no. 1, p. 460, 2023, doi: 10.3390/s23010460.

BIOGRAPHIES OF AUTHORS

Abdelmadjid Guessoum Graba    received the engineering degree in computer science from the University of Sidi Bel Abbes, Algeria, in 2006, the master's degree in Computer Science in 2010, and the Ph.D. degree in 2024, all from the Department of Computer Science, University of Sidi Bel Abbes, Algeria. He is currently a lecturer in the Telecommunications Department at the Faculty of Electrical Engineering, University of Sidi Bel Abbes, Algeria. His research interests include data mining, big data, artificial intelligence, and computer networks. He can be contacted at email: abdelmadjid.graba@univ-sba.dz.



Djoher Dalila Graba    received the master's degree in computer science in 2015 from the Department of Computer Science, University of Sidi Bel Abbes, Algeria, and the Ph.D. degree from the High School of Computer Engineering, Sidi Bel Abbes, Algeria, in 2022. She worked at the National Company for Electronic Industries (ENIE) for ten years. She is currently a lecturer in the Department of Mathematics at the University of Sidi Bel Abbes, Algeria, since 2024. Her research interests include software engineering, programming languages, and artificial intelligence. She can be contacted at email: dalila.graba@univ-sba.dz.